Typed Alignment and Audit Signatures

Realization-Relative Comparison for SOF Reports

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This paper is Paper XIII of the RIME program. It consumes versioned SOF Reports from Paper XII and owns explicit semantic alignment, partial comparability, typed Audit Signatures, and fixed-frame structural pseudodistance. A report pair is input evidence, not yet a comparison object.

Abstract

Problem. Paper XII defines a family of realization- and profile-relative single-system report views $\mathcal{R}^{\text{view}}_{\sigma,\eta,P_X,P_A,v_N,v_A}$, not a canonical map from a source system to one report. Two reports are therefore not comparable merely because they conform to the same reporting protocol. Statements such as "support decreased" remain undefined until record kinds, declared capabilities, sectors, observables, thresholds, depth conventions, and parameter samples have been admitted and aligned. This paper asks when two coordinates in a common typed language actually have shared semantics, when comparison must remain partial, and when it is formally incomparable.
Approach. This paper inherits carrier, policy, evidence, record-kind, and promotion guards from the Paper X compiler contracts. Its new alignment datum is

$$\mathfrak{A}_{\text{align}} = (\mathcal{R}^\star, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}).$$

Here $\mathcal{R}^\star$ and $\widehat{\mathcal{R}}$ are realization-relative reference and target reports. The inherited guards first retain only fields whose carriers, conventions, policies, evidence, and record-kind permissions are jointly eligible; a missing capability is not serialized as a zero. $\Phi_{\text{sec}}, \Phi_{\text{obs}}$ then align admitted sectors and observables in common retained coordinates. The resulting alignment-relative SOF comparison object is

$$\mathfrak{C}_{\text{cmp}} = (\mathfrak{A}_{\text{align}}, \Theta) = (\mathcal{R}^\star, \widehat{\mathcal{R}}, \Phi_{\text{sec}}, \Phi_{\text{obs}}; \Theta),$$

where Θ fixes normalization, thresholds, depth semantics, path synchronization, metrics, and aggregation. For an Audit Profile P_{audit}, comparison then induces the sparse typed map

$$\Delta^{P_{\text{audit}}}_{\text{audit}} = \{\Delta_\kappa\}_{\kappa \in \text{AlignCap}(\mathcal{R}^\star, \widehat{\mathcal{R}}; \Phi, \Theta, P_{\text{audit}})}.$$

The report pair alone is not comparable, the alignment datum alone does not fix comparison semantics, and the audit map is an output rather than the comparison object. Each requested coordinate carries a typed aligned, mismatched, unavailable, incomparable, or unresolved state.
Results. The **No Comparison Without Alignment** proposition states that a report pair does not induce an Audit Signature before both the alignment evidence Φ and comparison semantics Θ are fixed. The **Alignment-Relative Audit Faithfulness** invariant requires every emitted coordinate to retain its report-item bindings, carrier, policy, evidence, and provenance, or to record an explicit unmatched, incomparable, or unresolved state. On each fixed comparison frame of aligned structural representations, the structural signatures carry weighted Hamming pseudometrics; strictly positive weights give a metric after quotienting by equality of the retained structural signature. This is a fixed-frame structural pseudodistance, not a metric on a universal audit vector. The eight historical channels are retained as the Standard Regime-A Audit Profile

rather than imposed on every comparison. The controlled source corpus contains 20 F1–F4 snapshot payloads across GridWorld, SIR, Traffic, Compiler IR, and the additional Network Routing domain. These are retained as bounded source-addressed observations pending object-level revalidation. The fixed census is 20 F1–F4 source payloads plus 5 F5 compatibility payloads plus 3 transformation controls, hence 28 legacy/control cases, together with one separate native GridWorld F4 factual v2 audit: 29 SOFAUDIT v2 artifacts in the reported corpus. Five frozen F5 path payloads are retained as source-addressed compatibility observations, but their wall coordinates remain `UNRESOLVED` because the v1 audit payloads do not bind two Paper XI wall signatures. In the GridWorld F4 control, the Lie channel changes on 8 ordered pairs while the word channel remains unchanged. Three legitimate before/after transformations have nonzero raw signatures but zero residual contract violations. A Migration/Assembly Certificate upgrades 28 frozen `.sofaudit` v1 records to the v2 sparse-profile contract, including typed unavailability and explicit cutoff normalization; it does not recertify their source-level comparison values. A separate native-v2 GridWorld F4 audit binds two native SOFRS v2 reports, explicit sector and observable alignments, a complete comparison specification, report-item bindings, and an independent oracle. Its Object Certificate recomputes the selected finite support, word, and Lie coordinates from frozen source matrices and obtains factual states `ALIGNED`, `ALIGNED`, and `MISMATCH` with mismatch counts $0, 0, 8$. The migrated F4 audit remains unresolved.
Implications. This paper introduces alignment-relative Audit Signatures, not a generic report-diff protocol. Comparison is undefined before inherited guards, explicit alignment evidence, and comparison semantics; after all three, report differences become structured coordinates rather than aggregate scores. A nonzero signature records change, not defect. Diagnostic analogues remain descriptor-level unless an explicit comparison map is declared. Interpretation and policy-relative disposition belong to Paper XIV, and cross-frame transport remains open.

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1 Introduction

Papers VIII–X establish the static typed object language, typed dynamic fields, and capability-aware compilation theory; Paper XI supplies sparse within-event wall records and morphology profiles [4, 3, 2, 6]. Paper XII then instantiates the Paper X compiler contracts as a versioned single-system reporting protocol [5]. Its report view is relative to a frozen source snapshot σ, an admitted adapter η, a Paper X Compiler Profile P_X, a Paper XII Assembly Profile P_A, and normative and assembly version closures v_N, v_A:

$$S \longrightarrow \mathrm{Adapter}_\eta \longrightarrow (M_\eta, I_\eta) \xrightarrow{\mathrm{Compile}_{v1}(\cdot,P_X)} \mathcal{O}_{\eta,P_X} \xrightarrow{\mathrm{Assemble}_{v2}(\cdot,P_A)} \mathcal{R}^{\mathrm{view}}_{\sigma,\eta,P_X,P_A,v_N,v_A}(S).$$

This paper shifts the focus from single-system reporting to cross-report comparison. A single report answers: *under this admitted adapter, capability declaration, policy set, and profile, what structure was reported?* Paper XII establishes that a single-system report is realization-relative and is not, by itself, a comparative correctness certificate. Two report files are therefore not yet a mathematical comparison object.

1.1 From Report Relativity to Report Alignment

Let the reference and target reports be

$$\mathcal{R}^\star = \mathcal{R}^{\mathrm{view}}_{\sigma^\star,\eta^\star,P^\star_X,P^\star_A,v^\star_N,v^\star_A}(S^\star), \qquad \widehat{\mathcal{R}} = \mathcal{R}^{\mathrm{view}}_{\widehat{\sigma},\widehat{\eta},\widehat{P}_X,\widehat{P}_A,\widehat{v}_N,\widehat{v}_A}(\widehat{S}).$$

When no ambiguity arises, the report indices are suppressed after this definition. In particular, this paper never folds P_A back into $\mathrm{Compile}_{v1}$.

A literal report difference may arise from the source systems, from the adapters, from the enabled capabilities, from the selected profiles, or from several of these at once. This paper does not infer which source-level cause produced a report difference. Instead, it requires the referenced reports to retain their respective provenance and requires the comparison object to declare which report-level fields are eligible and how they are aligned.

Report relativity does not make comparison impossible. It identifies the additional data that comparison must not hide. In particular, alignment cannot repair an undeclared carrier, reinterpret `NOT_DECLARED` as zero, turn `UNREACHED_AT_CUTOFF` into infinity, or promote a diagnostic analogue into a strict SOF theorem instance.

1.2 Inherited Compiler Guards

This paper does not define a third admission system. It inherits record kinds, carrier declarations, policy and evidence requirements, unavailable-state semantics, and promotion rules from Paper X and the SOFRS v2 reports of Paper XII. Its new datum is the explicit pairwise alignment $\Phi = (\Phi_{\mathrm{sec}}, \Phi_{\mathrm{obs}})$ together with the comparison specification Θ.

Only compatible strict fields are compared. Analogue fields remain at the descriptor level. A comparison containing one strict and one analogue report requires an explicitly declared common shadow. This does not convert the analogue into a strict carrier or define a new record kind. The `.sofaudit` v2 validator checks inherited carrier, policy, evidence, record-kind, and promotion conditions before emitting any coordinate.

Before alignment, even a basic statement such as "the target has less support" has no invariant meaning. The apparent difference may arise because:

- the reports have incompatible record kinds or capability declarations;
- the two reports use different sectors;
- their observable families are not in correspondence;
- their thresholds or depth semantics differ;
- their parameter samples or trajectory samples are not synchronized.

The pair $(\mathcal{R}^\star, \widehat{\mathcal{R}})$ is therefore insufficient. The alignment datum

$$\mathfrak{A}_{\mathrm{align}} = (\mathcal{R}^\star, \widehat{\mathcal{R}}, \Phi_{\mathrm{sec}}, \Phi_{\mathrm{obs}})$$

places the reports in common retained coordinates. The additional specification Θ fixes how those coordinates are compared. Together they form the SOF comparison object

$$\mathfrak{C}_{\mathrm{cmp}} = (\mathfrak{A}_{\mathrm{align}}, \Theta).$$

> **SOF Report Alignment Principle.** Compare a target report $\widehat{\mathcal{R}}$ against a reference report $\mathcal{R}^\star$ only after sector and observable alignments Φ_{sec} and Φ_{obs} are explicit. The admissible comparison object is $(\mathfrak{A}_{\mathrm{align}}, \Theta)$; Δ_{audit} is its induced comparison signature.

Proposition 1.1 (No Comparison Without Alignment)**.**
Within the present object language there is no admissible map

$$(\mathcal{R}_1, \mathcal{R}_2) \longmapsto \Delta_{\mathrm{audit}}$$

from a bare report pair. An Audit Signature is defined only after explicit Φ identifies common retained semantic coordinates and Θ fixes the comparison policies on those coordinates.

Argument. *Each report is indexed by its own realization, carrier labels, policies, and evidence. Before* Φ*, equal field names do not establish a common comparison object; before* Θ*, even aligned values lack fixed normalization, cutoff, orientation, and mismatch semantics. Therefore a bare pair can support only a request for alignment, not a factual Audit Signature. This is an object-domain restriction, not merely a program branch that refuses to run.*

Protocol Invariant 1 (Alignment-Relative Audit Faithfulness). *Relative to a declared alignment contract* (Φ, Θ)*, every emitted Audit Signature coordinate must bind the contributing reference and target report items, retain their carrier, policy, evidence, and provenance qualifications, or record an explicit unmatched, incomparable, or unresolved state. Assembly must not manufacture a missing value, treat the reference role as truth status, or promote an unavailable coordinate to numerical zero.*

This invariant is protocol-level. It does not establish that Φ *is the scientifically best alignment or that either report is correct about its source.*

This paper makes four contributions:

1. *It defines the SOF Report Alignment datum and the typed alignment-relative comparison object* $\mathfrak{C}_{\mathrm{cmp}} = (\mathfrak{A}_{\mathrm{align}}, \Theta)$*, making comparability explicit rather than hidden preprocessing.*

2. *It defines capability-aware Audit Profiles and sparse typed comparison maps; unavailable coordinates remain explicit states rather than numerical zeros.*

3. *It defines a common retained comparison frame and equips representations embedded in that frame with a weighted structural pseudometric.*

4. *It separates protocol and migration conformance from a selected first-principles Object Certificate. A native-v2 GridWorld F4 audit closes the object-to-report-to-comparison chain, while the remaining F1–F4 payloads in GridWorld, SIR, Traffic, Compiler IR, and Network Routing remain bounded source observations. The corpus also includes legitimate transformations whose nonzero differences are licensed by contract, and supplies a versioned* `.sofaudit` *migration certificate that preserves five legacy F5 paths without promoting them to wall comparisons.*

The theory developed here is local. The paper does not define comparison between different retained frames, infer alignments automatically, prove a metric on a universal audit map, mutate the frozen `.sofaudit` v1 corpus, or turn difference into defect or action. The v2 corpus is a source-addressed migration rather than a historical rewrite. Single-report production belongs to Paper XII, aligned comparison to Paper XIII, and context-indexed interpretation and action semantics to Paper XIV.

The paper proceeds from the comparison object to the induced audit signature, its fixed-frame structural pseudodistance, and word/Lie separation. It then presents the native object control, bounded source observations, portability analysis, and failure boundaries. The appendices state the machine-readable alignment contract, reproducibility map, and geometric boundary; broader alignment regimes are deferred to the Outlook.

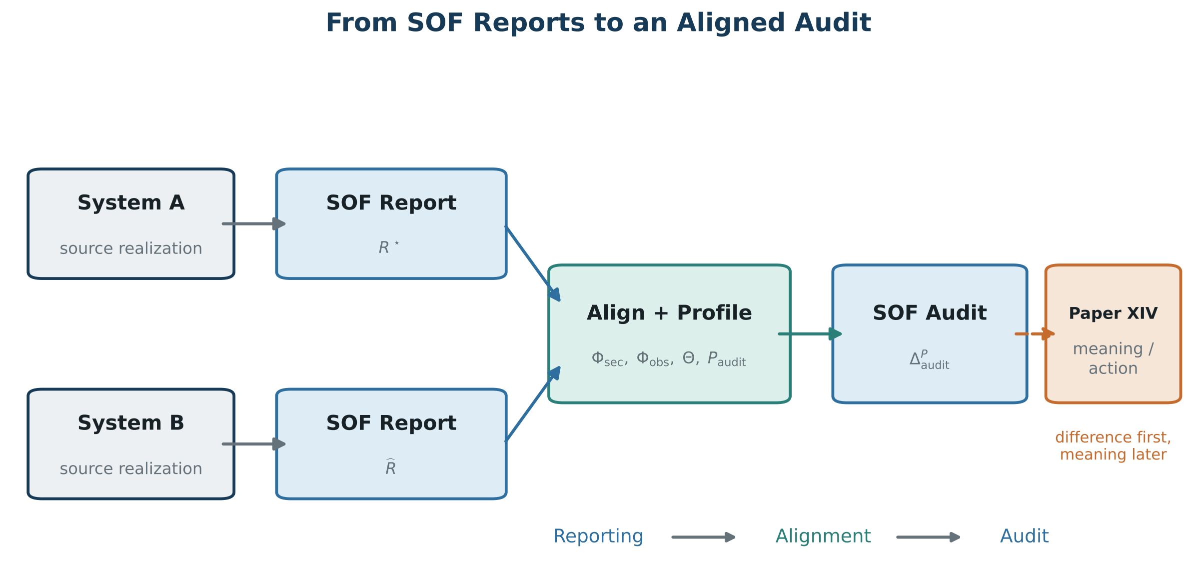

Figure 1. From SOF Reports to an audit signature. Two independently generated SOF Reports, their explicit alignment, and the comparison specification form $\mathfrak{C}_{\mathrm{cmp}}$. The induced Δ_{audit} records difference; interpretation and Action Semantics remain downstream.

2 Related Work and Novelty Boundary

Program lineage. Paper VIII supplies the SOF object layer; Paper IX supplies typed dynamic fields; Paper X supplies capability-sound compiler contracts and Registry evidence; Paper XI supplies typed one-wall records; and Paper XII supplies the versioned single-report protocol and alignment-ready provenance consumed here [4, 3, 2, 6, 5]. Cross-paper consumption does not transfer ownership: this paper owns aligned pairwise differences, not either input's admission, morphology, or report semantics.

Reporting and audit precedents. Model cards and end-to-end algorithmic audit frameworks provide precedents for structured records with explicit scope and failure boundaries [14, 16]. This paper differs by treating sector and observable alignment as the mathematical prerequisite for a comparison signature.

Schema and ontology alignment. Database schema matching and ontology matching construct correspondences between fields, entities, and semantic relations [15, 7]. SOF Report Alignment has a different comparison unit: sector correspondences and observable correspondences must be declared together with threshold, depth, trajectory, and claim-status semantics before a comparison signature exists.

Representation matching. Orthogonal Procrustes alignment and modern representation-similarity methods compare coordinate representations or learned feature spaces [17, 11]. This paper does not compare raw embeddings. It aligns declared observable structures and then compares their support, bridge, depth, frozen, response, constraint, and admitted wall-record coordinates.

Structured metrics. Hamming distance and graph edit distance provide standard precedents for discrete structural comparison [10, 8]. The fixed-frame pseudometric here is more restricted than a general report distance: it is defined only on aligned structural representations under fixed (Φ, Θ), and no metric is claimed on the full directional and path-dependent audit signature.

World-model comparison context. Latent-state world models and executable consequence benchmarks motivate the Regime B/C comparison problem [12, 9, 1, 13]. They are comparison contexts, not evidence for the fixed-frame proposition or the GridWorld F4 observation.

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3 SOF Report Alignment

3.1 Alignment Datum and Comparison Object

For a report $\mathcal{R}$, write $I(\mathcal{R})$ for its retained sector-label set and $G(\mathcal{R})$ for its retained observable-label set. An alignment places the two reports in common retained coordinate sets I_Φ and G_Φ. Schematically, its sector component is the span

$$I(\mathcal{R}^\star)_{\mathrm{ret}} \xrightarrow{\phi^\star_{\mathrm{sec}}} I_\Phi \xleftarrow{\widehat{\phi}_{\mathrm{sec}}} I(\widehat{\mathcal{R}})_{\mathrm{ret}},$$

and its observable component is defined analogously over G_Φ. The legs may be identities, relabellings, or declared aggregations. They need not be total or bijective, but unmatched coordinates and any aggregation rule must be explicit. A declared aggregation is a typed alignment operation with a family-specific law recorded in Θ; it is not thereby a canonical presheaf restriction or an untyped coarse-graining default.

The SOF Report Alignment datum is

$$\mathfrak{A}_{\mathrm{align}} = (\mathcal{R}^\star, \widehat{\mathcal{R}}, \Phi_{\mathrm{sec}}, \Phi_{\mathrm{obs}}), \qquad \Phi = (\Phi_{\mathrm{sec}}, \Phi_{\mathrm{obs}}).$$

Definition 3.1 (SOF Report Alignment)**.**

A SOF Report Alignment consists of a reference report, a target report, a typed sector alignment, and a typed observable alignment into common retained coordinates. Each alignment declares its map kind, matched and unmatched identifiers, totality/injectivity/ surjectivity properties, semantic basis, evidence, and negative boundary. The reference role is not a truth claim: its role basis must state whether it is a formal specification, exact recomputation, certified measurement, consensus standard, or declared baseline only. It is the minimal geometric datum required before a domain-independent report comparison can be specified.

The comparison specification is written schematically as

$$\Theta = (\mathsf{N}, \mathsf{M}, \mathsf{S}_D, \tau, \mathsf{P}, \mathsf{A}),$$

where N specifies normalization and scaling, M specifies the comparison metric and its coordinate weights, S_D fixes depth semantics and frozen-value handling, τ records thresholds and tolerances, P synchronizes parameterized paths, and A specifies aggregation. Components fixed by a schema version or protocol default need not be repeated in every record, but they remain part of the declared comparison semantics.

Definition 3.2 (SOF comparison object)**.**

A SOF comparison object is

$$\mathfrak{C}_{\mathrm{cmp}} = (\mathfrak{A}_{\mathrm{align}}, \Theta) = (\mathcal{R}^\star, \widehat{\mathcal{R}}, \Phi_{\mathrm{sec}}, \Phi_{\mathrm{obs}}; \Theta).$$

It is **admissible** when the inherited Paper X record-kind, carrier, policy, evidence, and promotion guards are satisfied; the report references resolve; Φ induces shape-compatible common coordinates for every compared structural array, unmatched coordinates are declared, and the controlled components of Θ fix normalization, metric, missing-value, depth, threshold, parameter, and aggregation semantics for every non-null comparison coordinate. Path-dependent coordinates additionally require declared parameter synchronization. An Object Certificate requires a separate comparison basis with raw source, independent recomputation, oracle result, and audit result artifacts.

Let $\mathrm{Cap}(\mathscr{R})$ denote the typed fields licensed by a report's Capability Manifest, IR, evidence, and profile. For a comparison profile P_{audit}, define

$$\mathrm{AlignCap}(\mathscr{R}^\star, \widehat{\mathscr{R}}; \Phi, \Theta, P_{\text{audit}}) = \mathrm{Cap}(\mathscr{R}^\star) \cap_{\Phi,\Theta} \mathrm{Cap}(\widehat{\mathscr{R}}) \cap \mathrm{Req}(P_{\text{audit}}).$$

The decorated intersection means that a field survives only when its carrier, semantic convention, run policy, evidence status, and alignment are compatible under (Φ, Θ). Comparison is therefore profile-indexed:

$$\boxed{\Delta_{\text{audit}}^{P_{\text{audit}}} = \mathrm{Compare}_{\Theta, P_{\text{audit}}}(\mathfrak{A}_{\text{align}}) = \{\Delta_\kappa\}_{\kappa \in \mathrm{AlignCap}}}.$$

Thus $\Delta_{\text{audit}}^{P_{\text{audit}}}$ is a sparse typed map, not a universal fixed-length vector. A requested coordinate may be `ALIGNED`, `MISMATCH`, `NOT_DECLARED`, `NOT_APPLICABLE`, `INCOMPARABLE`, or `UNRESOLVED`. Here `ALIGNED` means that comparability has been established and the retained values are equal or within the declared tolerance; `MISMATCH` means that comparability has been established and the values are unequal or outside that tolerance. Thus these two states lie on one result axis. Unavailable states carry no numerical zero and no affirmative claim status. The serialized record retains the comparison object, Audit Profile, inherited guard checks, and coordinate results as distinct fields.

Remark 3.3 (Comparison-object non-uniqueness).

Alignment is not unique in general. Different admissible choices of Φ_{sec} or Φ_{obs} may produce different comparison signatures under the same Θ. Different admissible choices of Θ may also change the signature under the same alignment. Both Φ and Θ therefore remain inside the `.sofaudit` comparison object rather than hidden preprocessing.

3.2 The Alignment Contract

Without Φ_{sec} and Φ_{obs}, two reports are merely two unrelated descriptions. The alignment contract establishes comparability:

- **Sector alignment** Φ_{sec}: In the simplest case (same ambient space, same sector partition), this is the identity. When sector counts differ or sectorizations are produced by different pipelines (e.g., reference from simulation, target from clustering), Φ_{sec} must specify which reference sector each target sector corresponds to, or declare pairs incomparable.

- **Observable alignment** Φ_{obs}: When observable families differ in cardinality or semantics (e.g., reference has 4 directional controls, target has 4 but in a different order), Φ_{obs} specifies the correspondence. When observables are incommensurable (reference: physical forces; target: neural network activations), alignment is defined at the sector-block support level rather than the operator level.

In the controlled source observations below, both alignments are identity maps or explicitly declared block/observable correspondences. Latent and behavioral comparisons generally require non-trivial alignments.

Notation Table

Paper XII distinguishes a realized object, compiled report, and serialized artifact. This paper inherits that distinction and adds the aligned comparison object and its serialization:

Symbol	Meaning
$\mathcal{F}_\eta$	Realized typed object $(V, \{Q_i\}, Y; X, \mathcal{H}_{\text{Hall}})$, with the Lie/Hall enrichment optional and independently declared
$\mathcal{O}_{\eta,P_X}$	Paper X `CompilerOutput` produced by $\text{Compile}_{v1}(M_\eta, I_\eta, P_X)$
$\mathcal{R}^{\text{norm}}_{\sigma,\eta,P_X,v_N}$	Normative single-system report core
$\mathcal{R}^{\text{view}}_{\sigma,\eta,P_X,P_A,v_N,v_A}$	Paper XII assembled report view
$\mathcal{A}_{\sigma,\eta,P_X,P_A,v_N,v_A,v_S}$	Serialized `.sofreport` artifact
$\mathcal{R}^\star$	Reference report
$\widehat{\mathcal{R}}$	Target report
$\mathfrak{A}_{\text{align}}$	SOF Report Alignment datum
$\mathfrak{C}_{\text{cmp}}$	Admissible SOF comparison object $(\mathfrak{A}_{\text{align}}, \Theta)$
Θ	Comparison specification: normalization, metric, depth semantics, thresholds, parameter synchronization, and aggregation
P_{audit}	Audit Profile selecting requested comparison coordinates
$\mathfrak{B} = (I_B, G_B, \Theta)$	Common retained comparison frame
$\mathfrak{F}_B$	Structural representations admissibly embedded into $\mathfrak{B}$
$\sigma_B(\mathcal{R})$	Aligned structural representation of one report in $\mathfrak{B}$
d_B	Fixed-frame structural pseudometric
Δ^P_{audit}	Sparse typed comparison map induced by alignment, Θ, and the Audit Profile

The `.sofaudit` file serializes $\mathfrak{C}_{\text{cmp}}$, the Audit Profile, inherited guard checks, the induced sparse comparison map, and its factual claim boundary.

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4 Audit Profiles and Fixed-Frame Geometry

4.1 Capability-Aware Audit Object

The general comparison output is the sparse map $\Delta^{P_{\text{audit}}}_{\text{audit}}$. It contains only coordinates requested by the Audit Profile and resolved through the aligned capabilities of both reports. The profile does not manufacture a missing word, Lie, deformation, or proxy carrier.

The controlled corpus uses the **Standard Regime-A Audit Profile** P_{A8}. It requests eight coordinate groups:

$$\Delta^{P_{\text{A8}}}_{\text{audit}} = (\,\Delta_{\text{supp}},\ \Delta_{\text{brw}},\ \Delta_{\text{brl}},\ \Delta_{\text{dep}},\ \Delta_{\text{frz}},\ \Delta_{\text{cns}},\ \Delta_{\text{ctrl}},\ \Delta_{\text{wal}}\,)$$

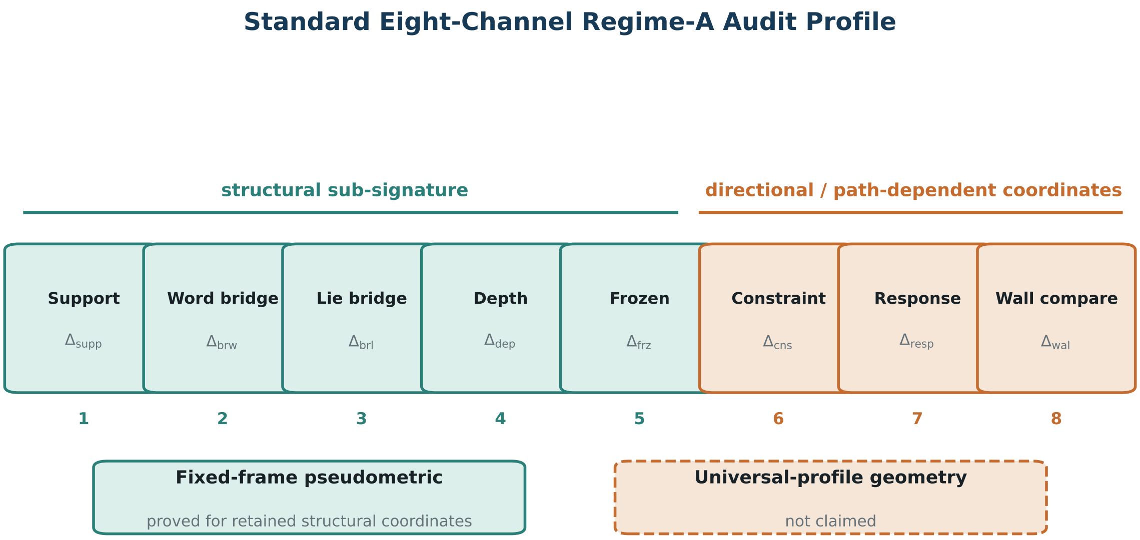

Figure 2. The Standard Regime-A Audit Profile. The first five requested coordinate groups form the structural sub-signature used by the fixed-frame pseudometric. Constraint, response, and wall-record coordinates retain directional or path-dependent semantics. No universal Audit Profile metric is claimed.

Dimension	Symbol	What is compared	What it detects
Support mismatch	Δ_{supp}	$R_1^\star$ vs. $\widehat{R}_1$	Missing or hallucinated direct transitions
Bridge mismatch (word)	Δ_{brw}	$R^\star_{2,\mathrm{word}}$ vs. $\widehat{R}_{2,\mathrm{word}}$	Lost or spurious 2-step word paths
Bridge mismatch (Lie)	Δ_{brl}	$R^\star_{2,\mathrm{Lie}}$ vs. $\widehat{R}_{2,\mathrm{Lie}}$	Commutator-structure changes
Depth distortion	Δ_{dep}	$D^\star_{\mathrm{word}}$ vs. $\widehat{D}_{\mathrm{word}}$	Path-length errors and pairwise frozen/reachable misclassification
Frozen disagreement	Δ_{frz}	$(f_{R_1}, f_{D,\mathrm{word}}, f_{D,\mathrm{Lie}})^\star$ vs. target counts	Net over/under-estimation of direct, word-depth, and Lie-depth reachability (three coordinates)
Constraint violations	Δ_{cns}	Raw $T^\star$ vs. target T	Target transitions absent from the reference
Control-response mismatch	Δ_{ctrl}	Per-control $c_a^\star(i,j)$ vs. $\hat{c}_a(i,j)$	Rate errors, native-control aliasing, response collapse
Wall-record mismatch	Δ_{wal}	Two retained Paper XI wall signatures in a compatible trajectory or domain context	Difference between aligned one-wall records

4.2 Standard Eight-Channel Regime-A Profile

The standard controlled profile has eight coordinate groups. Δ_{frz} is one group with three coordinates (`frozen_R1`, `frozen_D_word`, and `frozen_D_lie`), and Δ_{wal} is the eighth group. This coordinate is available only when both reports retain independently admitted one-wall records and Paper XI signatures.

4.3 Wall-Record Comparison Gate

Let $e^\star$ and $\hat{e}$ be wall atoms already admitted upstream by Paper IX and recorded by Paper XI. This paper consumes, but does not redefine, their one-record signatures

$$s^\star = \mathrm{MorphSig}_W^{P_W}(e^\star), \qquad \hat{s} = \mathrm{MorphSig}_W^{\widehat{P}_W}(\hat{e}).$$

The default comparison is curation-independent. Curated signatures may be compared only when the curation rulebook version, assignment semantics, and any override policy are also explicitly aligned; a shared tag spelling alone does not identify wall morphology.

An aligned wall-comparison input consists of the two source-addressed record and signature references together with a comparison context

$$\mathfrak{A}_{\text{wall}} = (s^\star, \hat{s}; \Psi_{\text{context}}, \Psi_{\text{field}}).$$

The context alignment declares whether the inputs are trajectory events or domain-level locus samples. For trajectories it also fixes parameter synchronization and orientation alignment; for locus samples it aligns the domain and incident-stratum semantics without imposing an intrinsic before/after order. The field alignment identifies the primary and retained context fields being compared. Only then may the Audit Profile emit

$$\Delta_{\text{wall}} = \mathrm{CompareWall}_\Theta(s^\star, \hat{s}; \Psi_{\text{context}}, \Psi_{\text{field}}).$$

This is a between-report difference. It is not either input's within-event δ_e^γ or within-locus δ_e^{loc}, and it cannot establish that either input was a wall. A path payload without two retained wall signatures is therefore `UNRESOLVED`, even when its sampled values differ.

The displayed coordinate Δ_{ctrl} concerns native system controls such as GridWorld moves, epidemiological rates, or traffic phases. It does not denote a downstream intervention action. SOFAUDIT v2 serializes it as the descriptive coordinate `response`; the frozen v1 records retain `action_response_failure` only as historical provenance.

The eight-channel profile is not universal. A report without a Lie/Hall carrier does not acquire a Lie-bridge value, and a static report does not acquire a wall record. Incompatible conventions and unresolved checks also remain nonnumeric typed states. None is numerical zero.

For the controlled tables below, $P_{\text{audit}} = P_{\text{A8}}$ is fixed and the superscript is suppressed. Every displayed dash in a wall column means `NOT_DECLARED`, not zero; `UNRESOLVED` marks a declared legacy path observation that lacks the wall-input binding required above. The dense tables abbreviate this state as `unres.`.

4.4 Fixed-Frame Structural Pseudodistance

Fix a common comparison frame

$$\mathfrak{B} = (I_B, G_B, \Theta),$$

The **fixed comparison frame** is the primary object. It is determined by the retained alignment and comparison specification and may be viewed locally as one fiber only in an informal sense; no global bundle or transition law between frames is claimed. Here I_B and G_B are the retained sector and observable index sets and Θ fixes their comparison semantics. Each admissible report carries a declared embedding $\phi_{\mathcal{R}} : \mathcal{R} \to \mathfrak{B}$. A pairwise alignment (Φ, Θ) is one way to construct such a frame by supplying the reference and target embeddings; it need not be the only pair that can be expressed in that frame. For an admissibly embedded report $\mathcal{R}$, let

$$\sigma_B(\mathcal{R}) = (R_1,\, R_{2,\text{word}},\, R_{2,\text{Lie}},\, D_{\text{word}},\, f_{R_1},\, f_{D,\text{word}},\, f_{D,\text{Lie}}).$$

This structural tuple is defined only on the Standard Regime-A profile subdomain where all listed operator, word, Lie/Hall, depth, and frozen-summary fields are jointly admitted. It is not synthesized for capability-sparse comparisons.

Here every matrix has first been reindexed or aggregated into the common coordinates of $\mathfrak{B}$ according to $\phi_{\mathcal{R}}$ and Θ. The first four coordinates are aligned matrices. The final three are the canonical frozen coordinates retained by Δ_{frz}. Write $\mathrm{Adm}(\mathfrak{B})$ for all reports with validated embeddings into this fixed frame.

Definition 4.1 (Fixed-frame structural representation set)**.**

The set $\mathfrak{F}_B$ is the collection of aligned structural representations

$$\mathfrak{F}_B = \{\sigma_B(\mathcal{R}) : \mathcal{R} \in \mathrm{Adm}(\mathfrak{B})\}.$$

Its elements are aligned structural representations, not unaligned SOF Reports and not audit deltas. Representations expressed under different retained frames belong to different sets. In a single pairwise audit, $\mathfrak{F}_B$ may contain only the reference and target representations; no larger population or manifold is implied.

For a matrix coordinate c, let H_c be off-diagonal entrywise Hamming distance; for a frozen-count coordinate, let H_c be the discrete distance. The weights $w_c \geq 0$ are the coordinate weights declared by Θ (part of M), not additional data introduced by the proposition. For $\sigma, \sigma' \in \mathfrak{F}_B$, set

$$d_B(\sigma, \sigma') = \sum_{c \in \mathcal{C}_{\mathrm{str}}} w_c \, H_c(\sigma_c, \sigma'_c).$$

Proposition 4.2 (Fixed-frame structural pseudometric)**.**

The function

$$d_B : \mathfrak{F}_B \times \mathfrak{F}_B \longrightarrow \mathbb{R}_{\geq 0}$$

is a pseudometric. If every $w_c > 0$, it is a metric on the retained aligned structural signature tuples, or equivalently on richer aligned representations quotiented by equality of those tuples. Omitted diagonal entries are part of the declared equivalence.

Proof. Every H_c is nonnegative and symmetric. Pointwise mismatch obeys

$$\mathbf{1}[x \neq z] \leq \mathbf{1}[x \neq y] + \mathbf{1}[y \neq z],$$

so each matrix Hamming coordinate satisfies the triangle inequality; the discrete frozen-count coordinates do as well. A nonnegative weighted sum preserves these properties. Zero-weight coordinates may identify distinct signatures, giving a pseudometric. If all weights are positive, zero distance is equivalent to equality of every retained structural coordinate.

This proposition applies only to the **structural sub-signature**. It does not establish a metric on the full Standard Regime-A audit map, still less on an arbitrary Audit Profile: constraint violations and control-response mismatches may be reference-to-target directional, while wall comparison requires a compatible trajectory or domain context. Those coordinates require separate symmetrization, normalization, and composition laws before a full-signature geometry can be claimed.

The fixed-frame statement is local. Each $\mathfrak{F}_B$ is a comparison domain in which retained coordinates and mismatch semantics have already been fixed. No transition law between distinct frames is defined; cross-frame geometry requires compatible pushforwards and coordinate-change laws.

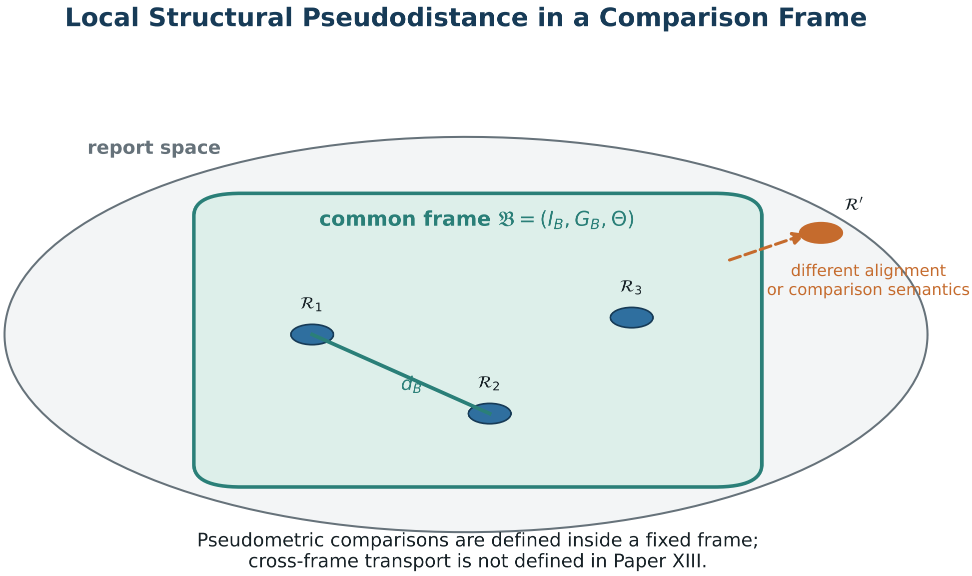

Figure 3. Alignment frames and local structural pseudodistance. Each frame fixes retained sector and observable coordinates together with the comparison semantics in Θ. The weighted structural pseudometric is defined within one comparison domain; the dashed cross-frame relation is intentionally left undefined.

4.5 Word Bridge vs. Lie Bridge

Paper VIII already establishes that positive word products and Lie/Hall commutators belong to different typed carriers. This paper does not reprove that static distinction. It requires the two coordinates to be aligned and compared separately.

For a matched pair of declared generators, the elementary identity

$$\|Q_i[X_a, X_b]Q_j\|_F \leq \|Q_i X_a X_b Q_j\|_F + \|Q_i X_b X_a Q_j\|_F$$

shows why Θ must align threshold conventions as well as labels: a thresholded commutator witness need not coincide with either ordered word-support shadow at the same absolute threshold. No coordinate is inferred from the other.

The native GridWorld F4 comparison is the paper-specific object-level control. On the tested skew-symmetrized generators, word products retain connectivity through reverse-direction components while the commutator channel changes. Hence $\Delta_{\mathrm{brw}} = 0$ while Δ_{brl} changes on eight ordered pairs. This is evidence for separate aligned coordinates, not a universal sensitivity ordering between word and Lie bridges.

4.6 First-Principles Object Control

The migrated `.sofaudit` cannot certify this matrix fact by schema validation or digest closure. The GridWorld F4 Object Certificate therefore starts from the two frozen native source snapshots, independently reconstructs the skew generator matrices, recomputes direct support, all ordered length-two products, and all simple commutators, and compares those results with both the native producer output and the frozen source payload. A graph-support baseline independently checks the direct transition incidence. Agreement among the source construction, matrix recomputation, and graph baseline is the object-level evidence. The certificate also checks that the migrated v2 audit leaves those coordinates `UNRESOLVED` rather than borrowing the source-level result. The separate native audit binds the oracle and may therefore emit factual coordinates; its SOFAUDIT validation receipt records protocol conformance separately.

The certificate is finite and scope-bounded. It establishes the selected F4 support/word/Lie counts and pair sets under the declared threshold and normalization. It does not establish that GridWorld is a scientifically adequate model of an external learned system or that the same sensitivity ordering holds in another realization.

4.7 Machine-Readable Comparison Record

Paired comparisons are serialized as `.sofaudit` records, distinct from the single-system `.sofreport` format of Paper XII. A `.sofaudit` record contains the declared alignment-relative comparison object $\mathfrak{C}_{\mathrm{cmp}} = (\mathcal{R}^{\star}, \widehat{\mathcal{R}}, \Phi; \Theta)$, inherited Paper X guard checks, an Audit Profile, the induced sparse `coordinates` map, and its factual claim boundary. It is a companion comparison contract rather than a revision of SOFRS. Appendix A states the v2 contract and Migration/Assembly Certificate; the frozen v1 artifacts remain source-addressed provenance.

Each positive coordinate carries both an evidence strength and an epistemic target. The `claim_target` field distinguishes external mathematical objects, empirical domain systems, representation interfaces, protocol conformance, and migration consistency; an ordinary SOFAUDIT claim targets the alignment-relative `comparison_relation`. A certificate additionally declares `certificate_class` as `object`, `comparison_audit`, `protocol_conformance`, or `migration_assembly`. The `comparison_audit` class establishes faithful calculation relative to declared Φ and Θ; it does not establish external-object truth. Schema validation, digest closure, and a conformance receipt cannot be relabeled as an Object Certificate. Object-level promotion additionally requires a bound comparison basis linking raw sources, independent recomputation, an oracle result, and the audit result.

4.8 Downstream Interpretation Boundary

The audit map records aligned disagreement. It does not assign severity, correctness, defect, or an intervention. Policy-relative interpretation and bounded candidate disposition belong to the downstream Paper XIV interpretation/action-semantics layer and cannot overwrite the recorded Paper XIII coordinates.

4.9 Legitimate Transformation Evidence

A nonzero comparison signature is descriptive, not condemnatory. For a declared legitimate transformation, an external source-addressed artifact may specify a transformation contract

$$\mathcal{T} = (\text{intent}, \text{allowed changes}, \text{preserved invariants}, \text{required postconditions}).$$

The raw Δ_{audit} remains unchanged. A separately typed contract evaluation may record residual changes not licensed by $\mathcal{T}$. Thus a large structural signature may still be conforming, while a small signature may violate a required invariant. SOFAUDIT v2 does not provide free-form `transformation_contract` or `contract_evaluation` escape fields; such evidence must be bound through a versioned external artifact and an admitted coordinate or future extension contract.

5 Controlled Source Observations: GridWorld

5.1 GridWorld Setup

The reference is a deterministic 5×5 grid with 25 one-hot cell sectors and an absorbing obstacle at $(2, 2)$. Moves originating at the obstacle self-loop, while neighboring moves into it are blocked. The observable family consists of the four skew-symmetrized directional transition matrices $\{N, S, E, W\}$. Five target perturbations test distinct comparison channels.

5.2 GridWorld Failure Modes

F1 – Action Aliasing (N = S). The target model's S transition matrix is replaced by N. Because the response constants retain block norms rather than orientation or sign, the support, bridge, depth, frozen, and response channels do not detect this opposite-action aliasing; only the raw transition-constraint channel responds. This exposes a boundary of the chosen observable family.

F2 – Persistence Loss. E from (1,1) is smeared across three targets (0.5, 0.3, 0.2 weights). The new edges lie on already-supported sector pairs, so direct and word support remain fixed while commutator paths change.

F3 – Forbidden Edge. S from (1,2) hallucinates a transition into the absorbing obstacle sector (2,2). This adds direct support, creates word and Lie bridges, shortens word depth, and violates the reference transition constraint.

F4 – Rare Bridge Deletion. E removed from all column-0 cells (0, 5, 10, 15, 20). Reverse-oriented skew components preserve the registered word paths, while the commutator channel and Lie-depth shadow change.

F5 – Obstacle-Path Compatibility Record. The frozen v1 payload sweeps obstacle position $(2, 2) \to (2, 1) \to (2, 0) \to (1, 0) \to (0, 0)$. The target remains fixed at obstacle=(2,1). The path records support, bridge, depth, and frozen-set changes; equal net frozen counts at some steps still conceal different frozen pairs. The payload does not bind two Paper XI wall signatures and declares no usable parameter synchronization, so SOFAUDIT v2 retains it as an unresolved path observation rather than a wall-record mismatch.

5.3 GridWorld Legacy Source-Payload Observations

The frozen F1–F5 vectors remain source-addressed in the legacy results indexed by `experiments/paper13/README.md`; they are not factual SOFAUDIT v2 coordinates. Migrated source-present values are `UNRESOLVED`, and absent values are `NOT_DECLARED`, until native report alignment and item bindings are supplied.

The five frozen controls have distinct source-payload activation patterns. In particular, the F4 payload records $\Delta_{\mathrm{brw}} = 0$ and $\Delta_{\mathrm{brl}} = 8$. This bounded observation motivates separate word and Lie channels but does not itself recertify a v2 comparison or imply a universal sensitivity ordering.

5.4 Native v2 End-to-End GridWorld F4 Validation

The native F4 control is a separate execution path, not an upgrade of the migrated record. It freezes two sparse GridWorld source snapshots, constructs two strict native SOFRS v2 reports with validation receipts, declares identity sector and observable alignments, fixes the complete comparison specification Θ, and binds each factual coordinate to report items. An independent validator reconstructs the matrices from the frozen sparse sources, computes direct generator support, ordered length-two word support, and simple commutator support without consuming producer caches, and checks direct support against the corresponding graph-incidence baseline.

The resulting native `.sofaudit` has `comparison_basis=COMPLETE` and binds the independent oracle. Its three requested factual coordinates are:

Native coordinate	State	Mismatch count
operator support	`ALIGNED`	0
ordered length-two word support	`ALIGNED`	0
simple-commutator support	`MISMATCH`	8

Thus the object-level result and the SOFAUDIT v2 factual comparison now form one digest-bound chain. The old migrated F4 artifact deliberately remains an `UNRESOLVED` preservation artifact; migration never borrows the new oracle. The native audit validation receipt certifies protocol conformance of the artifact closure, while the separately bound independent recomputation carries the Object Certificate.

---

6 Cross-Domain Controlled Source Observations

Each frozen domain producer supplies its own sectorization, observable family, and legacy comparison payload while retaining the same source-table grammar. No equivalence of the native dynamics or failure mechanisms is assumed. Except for the native GridWorld F4 chain above, these payloads remain bounded observations pending object-level revalidation and native-v2 item bindings.

6.1 SIR Compartmental Model

6.1.1 SIR Realization

The reference SIR system has three one-hot compartment sectors, Susceptible, Infectious, and Recovered, with rate observables β for $S \to I$ and γ for $I \to R$. The reference parameters are $\beta = 0.3$ and $\gamma = 0.1$, so $R_0 = 3$. The directed chain is $S \xrightarrow{\beta} I \xrightarrow{\gamma} R$. Under the skew-symmetrized rate generators:

- R_1: $S \leftrightarrow I$ and $I \leftrightarrow R$ each contribute 2 ordered off-diagonal pairs, yielding $R_1^{\text{offdiag}} = 4$. The skew part records antisymmetric support: $X_\beta[I,S] = \beta/2$ and $X_\beta[S,I] = -\beta/2$ together encode the S–I support channel. This is a proxy for the directed epidemiological $S \to I$ semantic, not a direct encoding.
- $R_{2,\text{word}}$: $S \leftrightarrow R$ via the 2-step word bridge $X_\gamma X_\beta$. $R_{2,\text{word}}^{\text{offdiag}} = 2$.
- D_{word}: $D(S,I) = 1$, $D(I,R) = 1$, and $D(S,R) = 2$, with symmetric reverse entries.
- $f_{R_1} = 2$ for the directly frozen S–R ordered pairs, while $f_{D,\text{word}} = f_{D,\text{Lie}} = 0$.
- $R_{2,\text{Lie}}$: $[X_\beta, X_\gamma]$ has a non-zero S–R block. $R_{2,\text{Lie}}^{\text{offdiag}} = 2$.

This is a minimal three-sector realization with a nontrivial two-step bridge.

6.1.2 SIR Failure Modes

F1 – Rate Equalization ($\beta = \gamma$). Binary structure is unchanged and only Δ_{ctrl} responds. The infection and recovery operators remain distinct because they occupy different support blocks.

F2 – Missing Edge ($\beta = 0$). The $S \to I$ channel vanishes, changing direct support, both bridge channels, depth, and all three frozen coordinates.

F3 – Forbidden Direct ($S \to R$). A spurious direct edge is added to the β observable, producing extra support and bridge paths together with one reference-constraint violation.

F4 – Rate Distortion ($\beta = 0.001$). Structure is preserved ($\beta > 0$ so edges exist), but the $S \to I$ response is 300 times weaker. Only the continuous control-response coordinate changes.

F5 – Beta-Trajectory Compatibility Record. β is swept from 0 to 0.5 in 11 steps; the target is fixed at $\beta = 0.2$. The frozen payload records sampled support and response changes but does not retain Paper XI wall signatures for both sides. The frozen-set mismatch occurs only at the endpoint $\beta = 0$; for all sampled $\beta > 0$, the reference and target have identical frozen sets.

6.1.3 SIR Legacy Source-Payload Observations

The complete frozen vectors remain in the source-addressed legacy results. The distinguishing bounded pattern is that binary coordinates change at $\beta = 0$, whereas for every sampled $\beta > 0$ the frozen-count delta is zero and rate changes remain continuous-response observations. These are not factual SOFAUDIT v2 coordinates.

6.1.4 Candidate Support Boundary and Response Crossing

The sampled SIR β-sweep displays one candidate source-side support boundary and one continuous response-order crossing:

- **$\beta = 0$ (candidate support boundary).** The $S \to I$ edge vanishes. R_1 loses the S–I support pair. The direct and depth-frozen counts jump. This paper records this sampled diagnostic but does not supply Paper IX wall admission or a Paper XI one-wall signature.
- **$R_0 = 1$, i.e., $\beta = \gamma$ (response-order crossing).** $\|X_\beta\|/\|X_\gamma\|$ crosses 1. No binary metric changes: R_1, R_2, and frozen sets are constant for all $\beta > 0$. The crossing is visible only in the continuous response constants.

In classical epidemiology, $R_0 = 1$ is already the critical epidemic threshold. SOF does not replace that interpretation; it re-expresses the same threshold as a crossing of registered response magnitudes rather than a change in compartmental support.

6.2 Traffic Intersection

The traffic-intersection control adds a third Regime A domain. The reference is a directed 2x2 intersection grid with four one-hot sectors (NW, NE, SW, SE) and two signal-phase observables. Phase A records N-S green movement through southbound edges NW $\to$ SW and NE $\to$ SE. Phase B records E–W green movement through westbound edges NE $\to$ NW and SE $\to$ SW. Reverse directions are represented by the skew-symmetric transpose, as in the other finite SOF controls.

The deformation parameter is the green-time ratio $\rho = t_A/t_B$. The reference is $\rho = 1$. Five constructed variants are compared:

- **F1 phase aliasing.** Phase B is replaced by Phase A, losing the horizontal response while making the two signal phases indistinguishable.

- **F2 missing phase.** $\rho = 0$, so Phase A is removed and vertical movement freezes.
- **F3 forbidden diagonal.** Phase A hallucinates an NW → SE diagonal edge.
- **F4 timing distortion.** $\rho = 0.01$, so binary support is preserved while control-response magnitudes collapse.
- **F5 rate-order trajectory.** The reference sweeps ρ from 0.01 to 100 while the target is fixed at $\rho = 2$.

6.2.1 Traffic Legacy Source-Payload Observations

The complete frozen vectors remain in the source-addressed legacy results and are not factual SOFAUDIT v2 coordinates. They separate phase aliasing, missing phase, forbidden-edge, and timing-response patterns without promoting the F5 path to a wall comparison.

Traffic F5 provides a rate-order trajectory diagnostic rather than binary wall evidence. The sampled interval $\rho \in [0.01, 100]$ excludes the limit points $\rho \to 0$ and $\rho \to \infty$, so frozen-count deltas remain $(0, 0, 0)$ under the declared tolerance. The record is therefore a trajectory-mismatch path, not a binary wall-crossing observation.

6.3 Compiler IR

The compiler-IR control uses two distinct observable families on five one-hot basic-block sectors:

$$(B_{0,\mathrm{entry}}, B_1, B_2, B_{3,\mathrm{side}}, B_{4,\mathrm{exit}}).$$

The observable family contains:

- X_{cfg}, recording control-flow edges;
- X_{defuse}, recording cross-block def-use chains.

The reference is an O2-like five-block IR where the side block B3 is reachable and the B3 → B4 data dependence is intact. Alignment is compiler-native: basic-block names, SSA provenance, CFG node correspondence, dominance metadata, or debug/source-span metadata can provide the comparison contract. The experiment uses identity alignment.

Five constructed variants are compared:

- **F1 CFG/def-use aliasing.** X_{defuse} is replaced by X_{cfg}, making data-flow structure indistinguishable from control flow.
- **F2 dead branch loss.** B3 is isolated by removing both CFG and def-use edges.
- **F3 spurious CFG edge.** A hallucinated B0 → B4 jump bypasses the normal path.
- **F4 lost def-use.** The B3 → B4 data edge is removed while the CFG edge is preserved.
- **F5 pass-pipeline path.** The reference follows O0 → mem2reg → simplifycfg, while the target remains fixed at the pre-simplifycfg snapshot.

6.3.1 Compiler IR Legacy Source-Payload Observations

The complete frozen vectors remain in the source-addressed legacy results and are not factual SOFAUDIT v2 coordinates. They retain separate CFG and def-use channels and keep the F5 pass-pipeline path unresolved as a wall comparison.

F4 is the cleanest compiler-specific diagnostic: aggregate support and word bridges remain unchanged, but the Lie bridge and per-channel response record the broken data-flow observable. F5 supplies a complementary legacy path observation: the reference crosses the `simplifycfg` transition while the target remains fixed at the pre-`simplifycfg` snapshot. Because neither side is bound to a Paper XI wall signature, this path difference is not emitted as a wall-record mismatch.

Compiler IR SOF Report Alignment is a structural comparison of control-flow and dependency preservation under declared alignment. It does not replace semantic equivalence checking or a verified compiler proof.

---

7 Portability and Failure Boundaries

7.1 Portability Criterion

A target domain follows the same construction when the required SOF realization and alignment contract can be made explicit:

1. identify a domain-justified sectorization;
2. identify the observable family and its semantics;
3. construct or designate a reference report;
4. construct or observe a target report;
5. form the admissible comparison object $\mathfrak{C}_{\mathrm{cmp}} = (\mathcal{R}^{\star}, \widehat{\mathcal{R}}, \Phi; \Theta)$;
6. compute the induced $\Delta_{\mathrm{audit}} = \mathrm{Compare}(\mathfrak{C}_{\mathrm{cmp}})$; and
7. record the comparison and its claim boundary.

7.2 Cross-Domain Synthesis

The four controlled domains isolate distinct roles for the same alignment grammar:

Domain	Sector origin	Observable family	Distinct alignment contribution
GridWorld	Cell sectors in a finite transition system	Directional action operators	Spatial support mismatch, obstacle-path diagnostics, and word/Lie bridge-channel contrast
SIR	Compartment sectors	Infection/recovery rate operators	Sampled support boundary versus smooth response-order crossing
Traffic	Intersection-node sectors	Directed signal-phase operators	Phase aliasing, timing distortion, and rate-order trajectory mismatch
Compiler IR	Basic-block sectors	CFG and def-use observables	Dual observable-family alignment and pass-pipeline path diagnostics

Their common comparison map is

$$\mathfrak{C}_{\text{cmp}} = (\mathcal{R}^{\star}, \widehat{\mathcal{R}}, \Phi; \Theta) \longmapsto \Delta_{\text{audit}} \longmapsto \text{typed diagnostic coordinates.}$$

The domains differ in sector origin and observable semantics, but the comparison-object type remains the same. The controlled comparisons therefore establish portability of the alignment grammar without identifying the underlying failure mechanisms across domains.

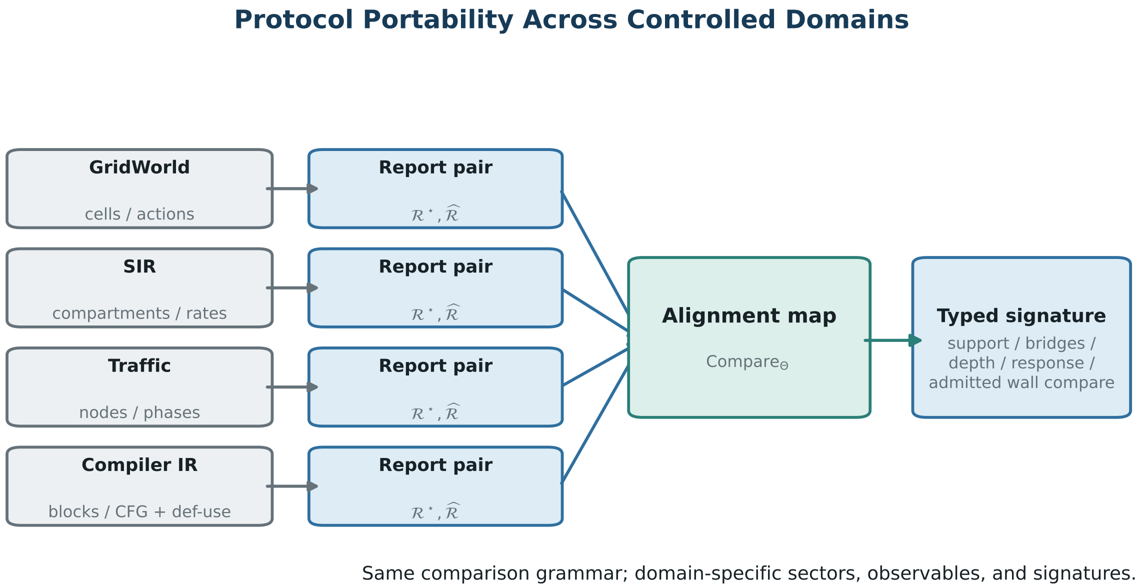

Figure 4. Protocol portability across controlled domains. GridWorld, SIR, Traffic, and Compiler IR use different sectors and observables, but each produces an explicit $\mathfrak{C}_{\text{cmp}}$ consumed by the same comparison map. The figure asserts portability of the comparison-object grammar, not equivalence of native dynamics.

7.3 Legitimate Transformation Controls

The failure-mode controls ask whether a target departs from a declared reference. The before/after controls ask a different question: what observable structure changes under a transformation already declared legitimate? Each control stores the full raw signature and a separate contract residual.

7.3.1 Legacy Source-Payload Transformation Observations

The source-addressed vectors cover a Compiler O0-like to O2-like transformation, Traffic $\rho : 0.5 \to 2.0$, and a GridWorld obstacle relocation. All three have zero declared contract residual despite nonzero raw differences; the complete legacy vectors remain outside the factual SOFAUDIT v2 coordinate surface.

The Compiler control uses a common five-block ambient realization with B3 marked as a retired target sector; its sector alignment is therefore explicit rather than silently treated as an unchanged active partition. Traffic preserves all binary topology while reversing the phase-response order. GridWorld changes support, bridges, depths, and frozen counts substantially, but every changed raw transition is incident to the declared old or new obstacle sector. In all three cases, nonzero Δ_{audit} means **changed**, while the zero contract residual means **no undeclared change was detected**.

7.4 Additional Validation Domain: Network Routing

Network Routing is included as an appendix validation domain rather than as a fifth member of the main synthesis table. It is useful because it isolates an ACL/policy-removal comparison role: the ACL acts as a constraint that removes edges from X_{external}, rather than an additive observable family like Compiler IR's CFG/def-use pair.

The reference has four prefix sectors ($P0_{\text{local}}, P1_{\text{dmz}}, P2_{\text{internal}}, P3_{\text{external}}$) and two observables: X_{internal} for trusted-zone routes and X_{external} for gateway routes.

7.4.1 Network Routing Legacy Source-Payload Observations

The complete frozen vectors remain in the source-addressed legacy results and are not factual SOFAUDIT v2 coordinates. The retained qualitative contrast is between F2 edge removal and F3 bridge/constraint activation.

Two signatures are especially useful. F2 is a pure ACL edge-removal pattern: direct support changes, but word and Lie bridge channels do not. F3 has no direct-support mismatch, yet it produces word/Lie bridge mismatches plus a constraint violation. This makes the appendix domain complementary to Compiler IR rather than another traffic-shaped graph example.

7.5 Failure Boundaries

Paper XII owns the single-report applicability boundaries. Paper XIII adds three paired-comparison boundaries:

- **Alignment failure.** If Φ_{sec} or Φ_{obs} cannot be specified (incommensurable sectorizations or observable families), the paired comparison reduces to two independent single-system reports. Undeclared coarse/fine refinement is an alignment failure. A refinement may be used inside one comparison object only when both legs into a common retained frame and the associated aggregation or pushforward rule are explicitly supplied in Φ and Θ. Transport between independently defined refinement frames remains open.
- **Coverage asymmetry.** If the reference covers state-space regions the target cannot represent (or vice versa), Δ_{frz} is dominated by coverage mismatch rather than a common-coordinate structural difference.
- **Comparison-resolution floor.** Thresholds, tolerances, sampling grids, and finite cutoffs bound the resolution of a coordinate. Near-zero rates may be indistinguishable from noise, threshold crossings may change discrete support, and unresolved finite depth remains `UNREACHED_AT_CUTOFF`.

7.6 Hostile Alignment Tests

Positive controlled comparisons do not test non-fabrication by themselves. The hostile suite rejects missing Φ or Θ, inconsistent carrier or policy guards, source-report receipt drift, analogue-to-strict promotion, unavailable-to-zero promotion, and unbound signature coordinates. Partial evidence must produce an unmatched, incomparable, or unresolved state rather than an invented coordinate.

A receipt cannot authenticate its own validator. A false `PASS` can be rejected only when the validator identity and implementation are bound to a trusted reference or when an independent validator reruns the checks. SOFAUDIT v2 does not consume the frozen v1 compatibility receipts: each legacy source report is first migrated through the Paper XII SOFRS v2 contract and receives a v2 validation receipt binding its CompilerOutput and assembly closure. That `PASS` still establishes protocol conformance rather than object-level truth.

A schema-valid but scientifically misleading alignment cannot be detected from schema shape alone. Such an alignment may receive protocol conformance but not an Object Certificate. Object-level promotion requires an independent domain baseline or a first-principles recomputation capable of falsifying the declared map.

The hostile suite also rejects free-form alignment objects, unconstrained metrics hidden inside Θ, an `ALIGNED` coordinate with a missing required alignment component, affirmative coordinates under rejected or unresolved compiler guards, and an external-object claim without a complete comparison oracle. A declared reference supports only difference from the selected baseline unless its role basis establishes stronger authority.

The cross-domain distance study is a separate computational boundary, not a fourth alignment failure class. It is a legacy source-payload diagnostic, not a factual SOFAUDIT v2 coordinate study. Across the 25 failure-mode comparisons, the normalized seven-coordinate structural vectors have same-label and different-label mean distances 1.0915 and 1.3991, a separation ratio of 1.2818. This is weak, domain-dominated separation. There is only one comparison per failure label within each domain, so no within-domain replication or clustering claim is made. This diagnostic uses only its declared shared fields; it does not promote the five F5 compatibility payloads to wall coordinates.

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8 Claim Spine

Evidence strength and claim target are orthogonal. Definitions and protocol invariants are not promoted into external scientific results.

Claim or object	Formal role and claim target	Reader-facing status
SOF comparison object $(\mathfrak{A}_{\text{align}}, \Theta)$ and Audit Profile	owned typed definitions; representation interface	not an independent evidence claim
No Comparison Without Alignment	domain-of-definition proposition; representation interface	Theorem
Alignment-Relative Audit Faithfulness	executable invariant; implementation checked by a Comparison Audit Certificate	not an independent evidence claim
Fixed-frame structural pseudometric	mathematical proposition under one retained frame and weights fixed by Θ	Theorem
GridWorld F4 first-principles oracle and native SOFAUDIT	finite end-to-end comparison; external-object target and Object Certificate	Computational Certificate
28-record v1-to-v2 conversion	finite preservation audit; Migration/Assembly Certificate	Computational Certificate
finite AB/BC/AC contextual-descent control	promoted bounded candidate-space witness only	Computational Certificate
remaining controlled F1–F4 legacy source payloads	source-payload diagnostics pending object-level revalidation; not factual v2 coordinates	Computational Observation
cross-frame transport, generalized descent, and inferred semantic alignment	open targets; no certificate is claimed here	Research Program

The `.sofaudit` validation receipt is a Protocol Conformance Certificate only. It is not evidence that either source report is scientifically adequate, that the declared alignment is uniquely correct, or that a nonzero coordinate is a defect.

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9 Conclusion

This paper fixes the SOF comparison object $\mathfrak{C}_{\mathrm{cmp}} = (\mathcal{R}^{\star}, \widehat{\mathcal{R}}, \Phi; \Theta)$. Alignment supplies common retained coordinates, Θ fixes their comparison semantics, and an Audit Profile selects the sparse typed output $\Delta_{\mathrm{audit}}^{P_{\mathrm{audit}}}$. In each fixed retained frame where the Standard Regime-A structural fields are jointly admitted, the structural sub-signature carries a weighted pseudometric.

The native GridWorld F4 chain demonstrates factual end-to-end comparison. The remaining GridWorld, SIR, Traffic, Compiler IR, Network Routing, and legitimate transformation payloads provide bounded cross-domain exercises of the comparison grammar. They do not extend the pseudometric to the full signature or define comparison between frames. Those extensions require normalization, symmetrization, and compatible pushforward laws.

Within the RIME program, Paper XII owns reporting. Paper XIII owns alignment-relative comparison and fixed-frame pseudodistance. Paper XIV owns context-indexed interpretation and policy-relative disposition. Their objects remain distinct: difference, interpretation, and bounded candidate generation.

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10 Outlook

10.1 Alignment Regimes

The SOF comparison object applies to three reference-target regimes:

Regime	Reference	Target	Alignment status
A: controlled reference	explicit dynamics or structure	constructed variant or declared transformation	validated here
B: aligned latent model	environment or analytic reference	latent-state dynamics model	requires a justified latent-sector correspondence
C: black-box behavioral	grounded observations	API-, trajectory-, or video-level model	requires semantic probe alignment

These regimes classify the relation between the two reports; they are independent of the SOF Applicability Hierarchy, which classifies the quality of each source-to-report realization. Regimes B and C remain future applications. World-model proposals and executable consequence benchmarks provide relevant contexts [9, 1, 13], but they do not supply evidence for the structural results proved here.

10.2 Comparison Tooling

A software implementation should preserve the mathematical order of operations: declare the report pair, record Φ and Θ, compute the signature, and only then apply domain-specific interpretation. Automatic alignment inference must remain distinguishable from a declared alignment, and trajectory comparison must retain parameter synchronization explicitly.

10.3 Presheaf and Descent Outlook

The fixed-frame boundary suggests a more structured Research Program. Let $\mathcal{C}_{\mathrm{obs}}$ be a candidate category of observation contexts, where a first-stage morphism

$$u : C' \longrightarrow C$$

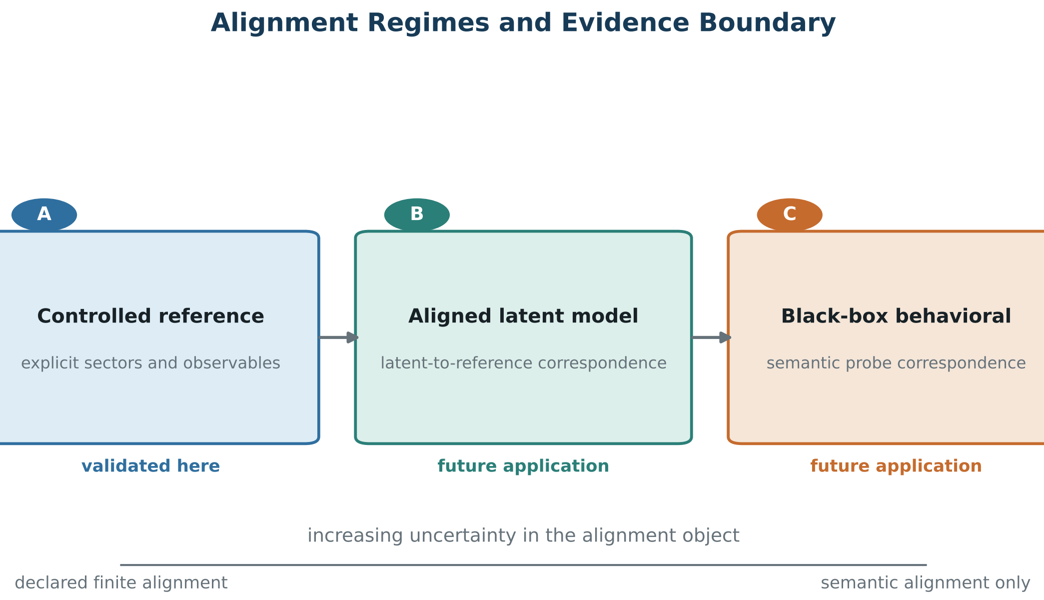

Figure 5. Alignment regimes and evidence boundary. Regime A supplies the controlled validation used in this paper. Regimes B and C share the comparison-object type but require increasingly strong latent-sector or semantic-probe justification; they remain application regimes rather than evidence for the fixed-frame results.

is restricted to localization, retained-sector or retained-observable restriction, parameter-domain restriction, or semantics-preserving relabelling for which a canonical restriction map exists. The exploratory formal prototype defines the finite typed partial-signature assignment

$$F_{\mathrm{sig}} : \mathcal{C}_{\mathrm{obs}}^{\mathrm{op}} \longrightarrow \mathbf{Set}.$$

All labels and convention packages in this prototype are drawn from fixed set-sized registries, so the first-stage context category is small.

Within the finite exploratory prototype, identity and composition are verified directly from the definition of coordinate restriction. This is an exact formal result internal to that prototype, not a general SOF presheaf theorem or a certificate claim of this paper. Presheaves of raw observables, full realizations, and comparison-ready records remain candidate extensions rather than results of this paper.

The carrier index is essential: restriction may preserve word data or Lie/Hall data, but it does not convert one carrier into the other. Likewise, a resolution refinement that splits a coarse sector does not automatically define a presheaf restriction, because a coarse relation need not determine its values on the finer sectors. Aggregation is instead a separate candidate pushforward and is available only when a family-specific law has been declared.

In the same-carrier, same-realization-kind, and same-convention subcase, a comparison defined here can factor through an explicit common retained context C^* with admissible canonical-restriction legs

$$C^* \longrightarrow C_{\mathrm{ref}}, \qquad C^* \longrightarrow C_{\mathrm{tar}},$$

so that both sections can be restricted to $F_{\mathrm{sig}}(C^*)$ before comparison. The term *common retained context* is intentional: C^* is not assumed to be a lattice-theoretic common refinement.

This span does not cover the full alignment contract defined here. In particular, a strict-vs-analogue comparison may be admitted by explicit Φ_{sec}, Φ_{obs}, and Θ, but it cannot be represented by two ordinary restriction legs because realization kind is preserved by every morphism of the first-stage context category. An alignment bridge is not thereby promoted to a presheaf restriction.

Finite gluing tests can then ask separately whether compatible local sections have a global extension and whether that extension is unique. The exploratory controls show why this is nontrivial for relation-valued signatures: sector subsets may cover all sector labels while failing to cover cross-subset relations, leaving multiple global extensions. This is a finite-candidate analogue of a separatedness failure, not a general separatedness theorem. Conversely, a bounded search can establish only that no global candidate was found in the declared candidate space; it cannot establish nonexistence in an unbounded presheaf. No Grothendieck topology, sheaf theorem, descent theorem, or topos-level structural result is claimed in this paper. The finite prototype remains non-evidentiary Research Program material; only the separately registered AB/BC/AC control in Appendix B is on the paper's evidence surface.

10.4 Open Problems

1. **Context category and descent.** Generalize the finite canonical-restriction prototype to a structurally natural observation-context category; determine which carrier-typed assignments remain functorial, identify natural covers, test their closure properties, and study existence and uniqueness of gluing.

2. **Cross-frame transport.** Define family-specific pushforward and transition laws for resolution refinement, aggregation, and changing observable families without treating them as automatic restrictions.

3. **Alignment inference and stability.** Determine when Φ_{sec} and Φ_{obs} can be inferred, and how latent clustering or quantization uncertainty propagates to Δ_{audit}.

4. **Full-signature geometry.** Extend the fixed-frame pseudometric, if possible, to directional constraints, response coordinates, and synchronized wall records, with normalization across sector counts.

5. **Observable sufficiency.** Characterize minimal observable families that preserve diagnostically relevant distinctions and support causal attribution of response mismatches.

6. **Directed comparison costs.** Determine whether normalized channel strengths define canonical sector-pair costs without model-dependent weighting choices.

Appendix A Alignment Artifact and Schema Validation

The canonical machine-readable contract is `schemas/sofaudit/v2.0.schema.json`. A conforming `.sofaudit` record encodes an aligned comparison, not a second single-system SOF Report and not a deployment decision. The separate `schemas/sofaudit/validation-receipt-v2.0.schema.json` binds one audit to the validator and digest-closed artifacts that were checked; the receipt is a Protocol Conformance Certificate, not object-level evidence.

The contract's conceptual closure is the comparison object, its explicit alignment and comparison specification, a versioned Audit Profile, sparse typed coordinates, comparison-basis evidence, and a negative action boundary. The full field inventory and shared reference types are defined by the source-addressed schema and common contracts, not copied into this appendix. In particular, `source_reports`, `alignment`, and `comparison_specification` serialize $\mathfrak{C}_{\mathrm{cmp}}$; `audit_` `profile` and `coordinates` serialize $\Delta_{\text{audit}}^{P_{\text{audit}}}$. A nonzero mismatch does not imply defect. `NOT_DECLARED`, `NOT_APPLICABLE`, `INCOMPARABLE`, and `UNRESOLVED` coordinates have null values and cannot be replaced by zero. Interpretation and action fields belong to Paper XIV.

The schema fixes object shapes, while the semantic validator recomputes their relations. It verifies reference and target roles, derives the regime from the two report kinds, closes the Audit Profile against the coordinate keys, recomputes alignment coverage and total/injective/surjective properties from the declared pair maps and report universes, and derives inherited guard state. It then enforces coupling among guards, alignment, and coordinates. It recomputes comparison-basis completeness, checks claim/result/certificate compatibility, and resolves every artifact identifier, role, and digest. An external-object claim is rejected unless its comparison basis binds an independent object oracle over raw source, independent recomputation, oracle result, and audit result artifacts. Hostile fixtures exercise these cross-field failures; they are not consequences of JSON Schema validation alone.

The frozen `schemas/sofaudit/v1.0.schema.json` and its 28 records remain immutable provenance. The migration adapter preserves their source payloads and unavailable states; it does not manufacture factual v2 `ALIGNED` or `MISMATCH` coordinates. The separate native GridWorld F4 chain binds two native SOFRS v2 reports, complete alignment and comparison semantics, and an independent oracle, while the migrated F4 artifact remains `UNRESOLVED`. The census, source/output digests, profile references, and receipt paths are indexed in `experiments/paper13/README.md`, `results/migration-index.json`, and the source-addressed `results/native/`, `results/object-certificates/`, and `results/receipts/` directories.

Those records distinguish Migration/Assembly Certificates from the native Object Certificate. No migration field or status is copied into the native chain, and no native oracle result is copied back into a migrated artifact.

Appendix B Computational Artifacts

The following source-addressed artifacts implement the v2 comparison contract and finite controls. Full inventories, digests, and generated outputs remain indexed in `experiments/paper13/README.md` rather than duplicated here.

Artifact	Role	Source-addressed path
B1	SOFAUDIT v2 object and receipt schemas	`schemas/sofaudit/v2.0.schema.json`; `schemas/sofaudit/validation-receipt-v2.0.schema.json`
B2	coordinate semantics registry and Standard Regime-A profile	`schemas/sofaudit/coordinate-semantics-registry-v1.0.json`; `schemas/sofaudit/paper13-standard-regime-a-profile-v2.0.json`
B3	semantic validator and receipt checks	`experiments/paper13/validation/validate_sofaudit_v2.py`
B4	v1-to-v2 migration adapter and index	`experiments/paper13/validation/migrate_sofaudit_v1_to_v2.py`; `experiments/paper13/results/migration-index.json`
B5	native GridWorld F4 producer and factual audit chain	`experiments/paper13/validation/gridworld_f4_native_v2.py`; `experiments/paper13/results/native/gridworld-f4/`
B6	independent GridWorld F4 object control	`experiments/paper13/validation/gridworld_f4_object_certificate.py`; `experiments/paper13/results/object-certificates/gridworld_f4.object-certificate.json`
B7	promoted finite AB/BC/AC control and validator	`experiments/paper13/validation/contextual_descent_control.py`; `experiments/paper13/validation/validate_contextual_descent_control.py`; `experiments/paper13/results/controls/`
B8	focused migration, native-chain, and hostile regressions	`tests/test_sofaudit_v2.py`

The artifact census is 20 F1–F4 source payloads, five F5 compatibility payloads, and three transformation controls, giving 28 migrated records, plus one separate native GridWorld F4 audit. This count is a corpus definition, not a generalization claim.

The result directories index the executable artifacts: `results/audits/` and `results/receipts/` hold migrated comparison records and receipts; `results/native/` and `results/object-certificates/` hold the factual native chain; and `results/controls/` holds the separately promoted finite descent control. The archived v1 tables and producers remain under `archive/` and are not factual v2 evidence.

B1–B8 establish only their declared finite targets. Migrated records are not factual v2 comparisons, and the AB/BC/AC control is not a general descent theorem. The artifact map and runnable contract checks are listed in `experiments/paper13/README.md`; all listed artifacts are available in the RIME repository.

Appendix C Geometric Boundary and Outlook

The computational controls establish fixed-frame and declared-threshold properties only. Sector permutations and conditional rescalings can be tested inside a declared frame, while threshold crossings can change the discrete signature and changes of alignment data move the representation to a different comparison domain.

The resulting local domains do not yet form a global comparison geometry. Transition maps and pushforward laws must be defined before bundle, connection, transport, or curvature language can be promoted from outlook to theorem.

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